



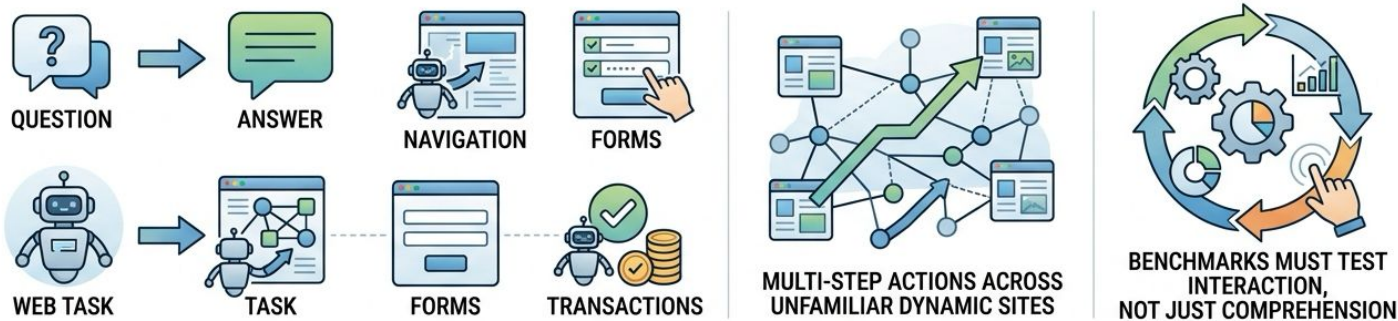
U.S. General Services
Administration

Chapter 3.3: Web Navigation and Multi-Hop Benchmarks

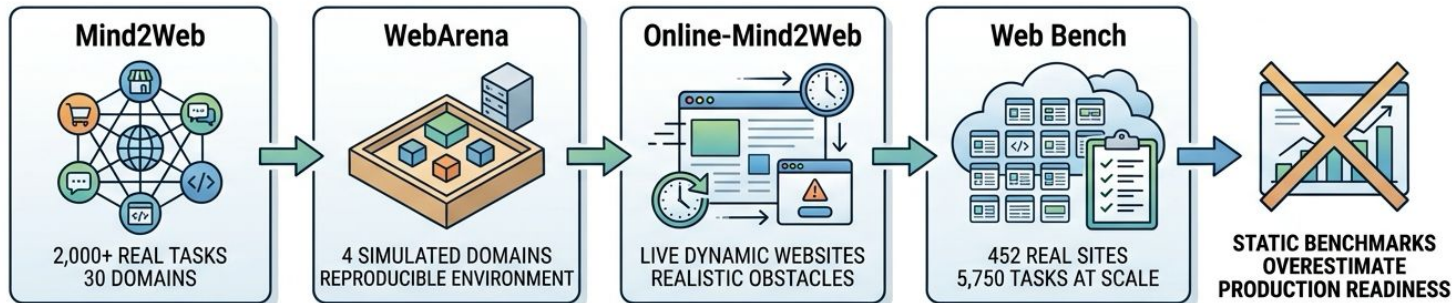
US General Service Administration
AI Community of Practice - March 2026

The Benchmark Evolution

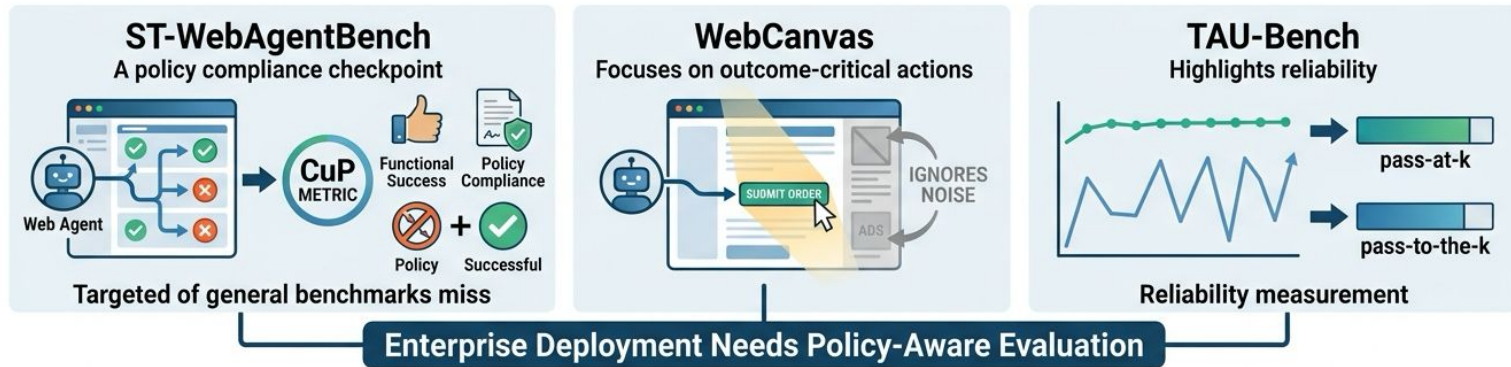
AGENTS BROWSING THE WEB



THE BENCHMARK EVOLUTION



Specialized Benchmarks Fill the Gaps

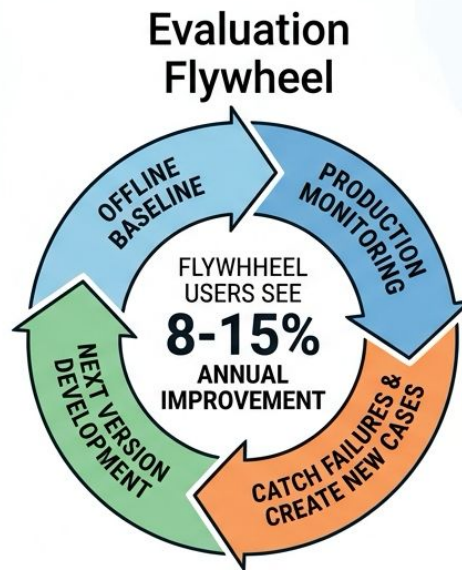
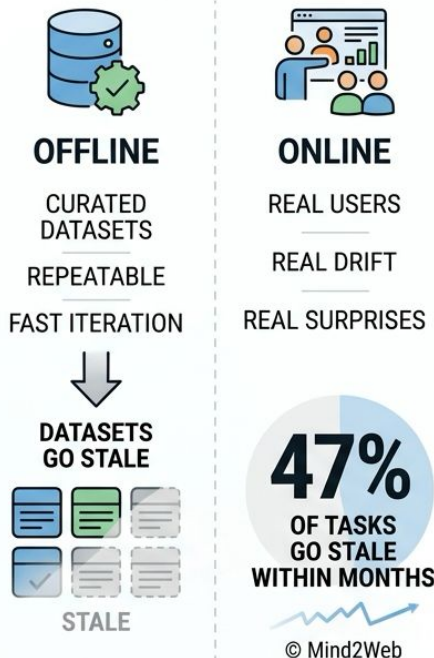


METRICS THAT MATTER FOR WEB AGENTS



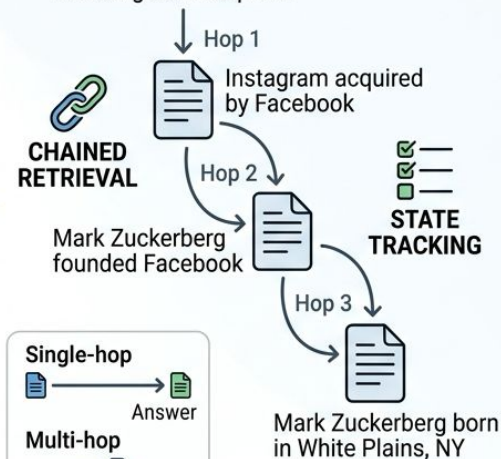
Offline vs. Online Evaluation

Offline vs. Online Evaluation

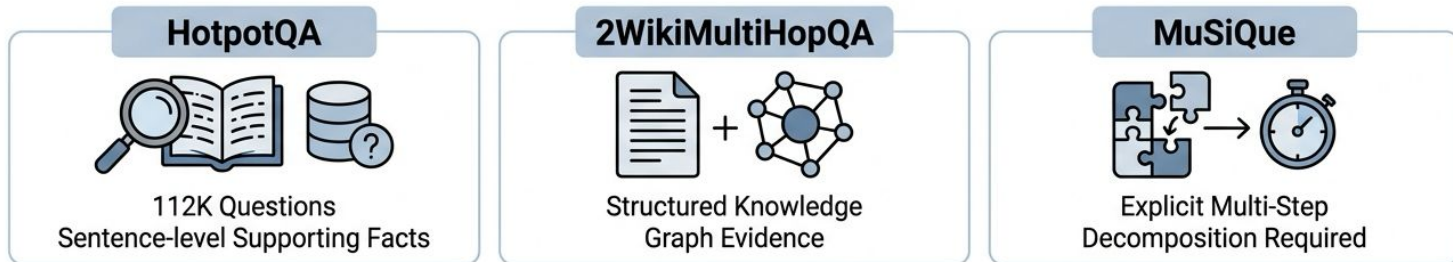


Multi-Hop Reasoning: A Different Challenge

Birthplace of the founder of Instagram's acquirer?



Multi-Hop Benchmarks and What They Reveal

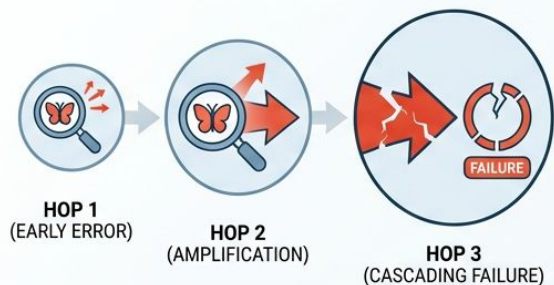


CRITICAL REVELATION: Most Correct Answers Lack Valid Reasoning Evidence

- Approx. 75% of Correct Answers Come Through Shortcuts or Hallucinations
- Not Derived from Sound Reasoning
- Dangerous for High-Stakes Applications (e.g., Healthcare, Finance, Legal)

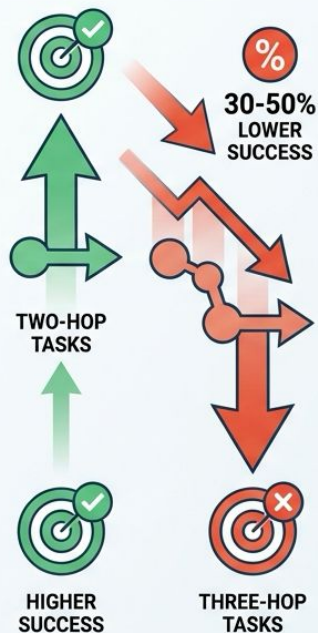
Error Propagation in Multi-Step Reasoning

THE BUTTERFLY EFFECT: CASCADING ERRORS



Early errors cascade through all subsequent hops

SUCCESS RATE DROP

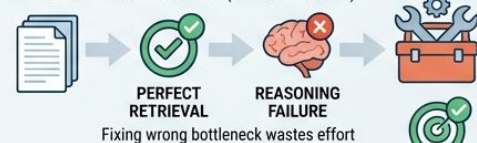


TWO FAILURE MODES



OPTIMIZATION & ANALYSIS

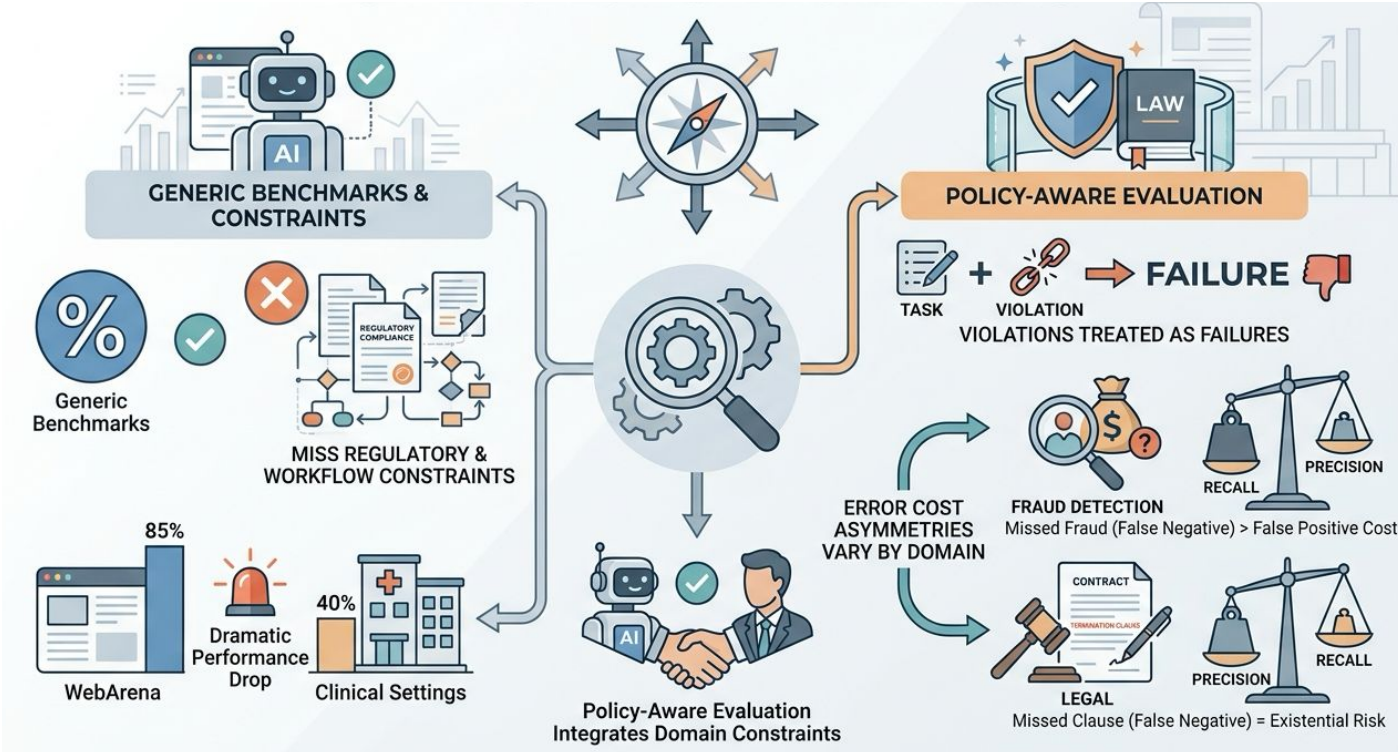
PATH A: WRONG BOTTLENECK (WASTED EFFORT)



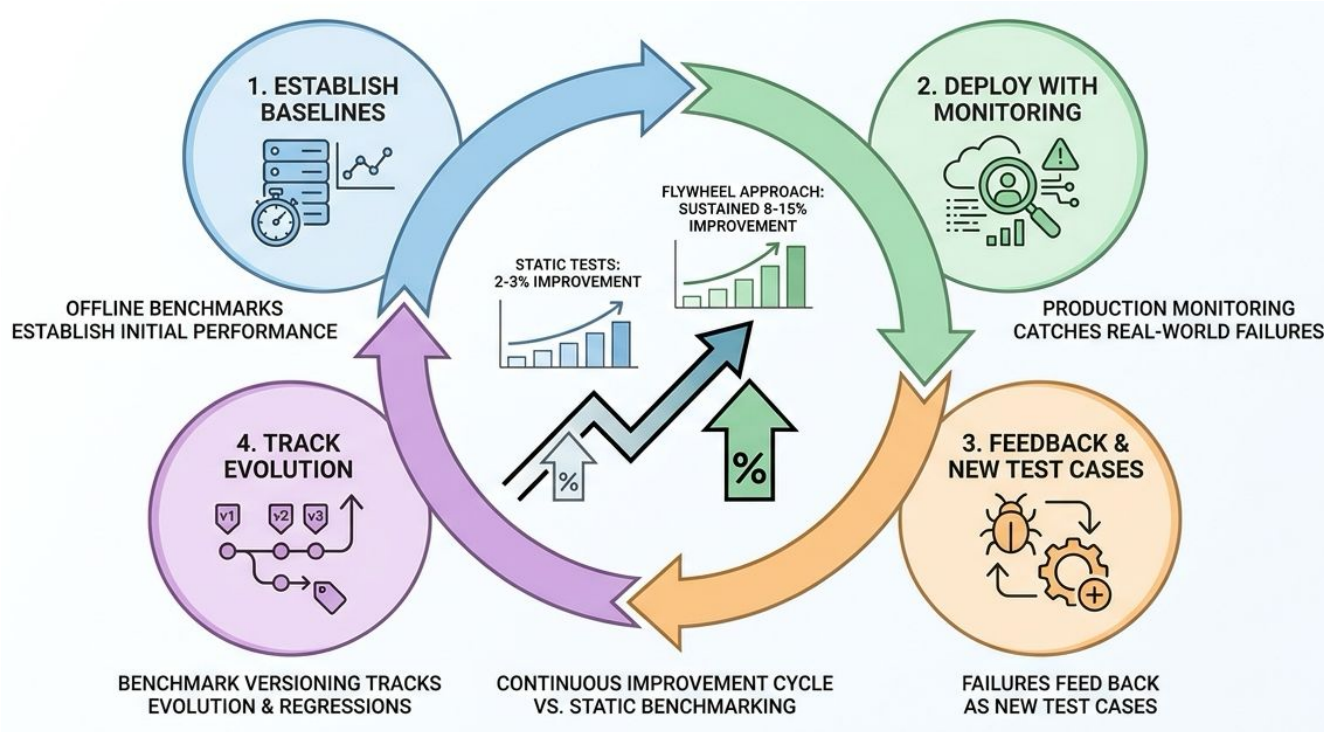
PATH B: STRATIFIED ANALYSIS (CORRECT BOTTLENECK)



Domain-Specific Benchmarking

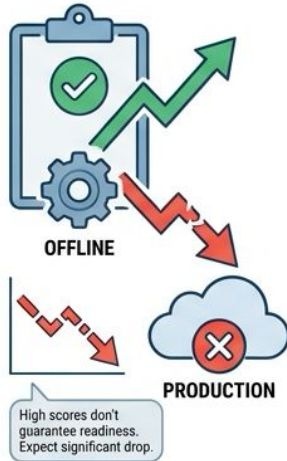


The Evaluation Flywheel in Practice



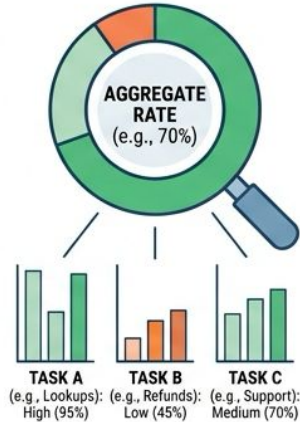
Common Misconceptions

OFFLINE SCORES VS. PRODUCTION SUCCESS



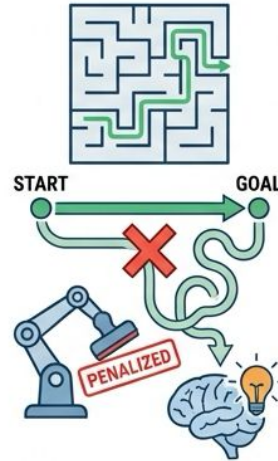
High scores don't guarantee readiness. Expect significant drop.

AGGREGATE SUCCESS MASKS PATTERNS



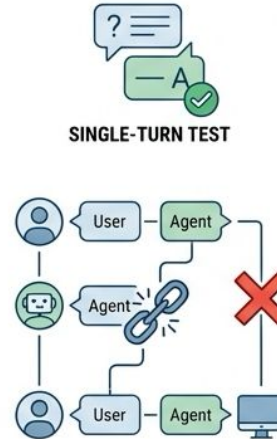
Success rate hides reliability and error clustering.

TRAJECTORY MATCHING PENALIZES CREATIVITY



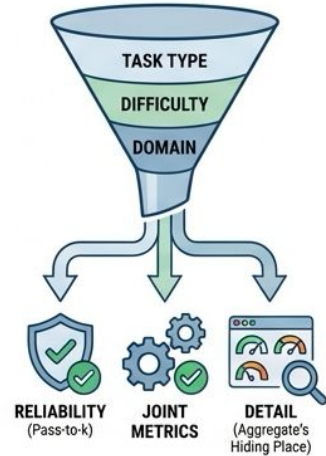
Valid alternative paths punished by rigid evaluation.

SINGLE-TURN TESTS MISS COMPLEXITY



Benchmarks fail to capture multi-turn conversation depth.

THE ANTIDOTE: STRATIFY EVERYTHING



Break down results by type, difficulty, and domain for true insight.

Key Takeaways

- Web and multi-hop benchmarks test complementary agent skills
 - No single metric captures readiness; use multidimensional evaluation
 - The evaluation flywheel is essential for sustained improvement
 - Domain-specific benchmarks catch what generic ones miss
 - Next: Chapter 3.4 on parameter tuning and optimization
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